ATE 2612Canine Infectious Diseases

CANINE DISTEMPER VIRUS

Pathogen

Paramyxovirus
Enveloped, ssRNA
Related to human measles

Transmission

Affects carnivores (dogs, raccoons, foxes, skunks, ferrets)
Aerosol droplets
Inhalation of airborne droplets
Direct contact with urine, feces, mucosal secretions

Pathophysiology

Initial replication in respiratory epithelium
Unknown mechanism, secondary bacteria involvement suspected
Multiplies in macrophages and lesser in GI
Spread to lymphoid tissue, CNS
Spreads through blood to epithelial cells throughout body
Immune response controls most infections, but causes immunopathogenic
Most viruses expelled, 1-2 weeks logs, persistent virus (days, CNS, GI tract)
No immune response / immune clinical signs, die acutely
Clinical signs over 7 days after exposure

Clinical Signs

- Severe respiratory signs: mucopurulent nasal discharge, conjunctivitis, cough, lethargy, dyspnea, fever
- Secondary bacteria
- CNS signs: jerking seizures, ataxia, hyperkinesias
- Neurovisceral
  - Hard footpads (hyperkeratosis)
  - Dental (out of enamel (enamel hypoplasia))
  - Dermal: hyperkeratosis of nose and footpads

Diagnostics

- CBC, biochemistry
- Cytology: inclusion bodies in cells (mainly lymphocytes)
- Antibody titers
  - Titer increase over 2 weeks post-vaccine
  - IgG or infection accounts are unreliable
  - **PCR on blood, conjunctival swabs, urine**

Treatment

- Supportive care
  - Supportive care
  - IV fluids
  - Antibiotics for secondary bacterial infection
  - Anticonvulsants
  - Antiemetics
  - Antitussives
  - Anti-inflammatories

Prognosis

- Depends on virulence and host immunity
  - Neurologic signs - ignore to poor
- Acute: 1 month after infection
- Chronic: 2 weeks - 3 months after infection

Prevention / Vaccination

- Routine distemper vaccination
  - Modified live virus (MLV) vaccine - DO
  - Often combined with other diseases
  - DAPP or DHPP
  - Not recommended immunosuppressed animals
- Titer - DO

Vaccination Schedule

- 16 days - 5 weeks
- Booster every 3 weeks until 16 - 18 weeks of age
- If series completed, DO annually rabies
- 1st dose at 8 weeks; 2 boosts 2 - 4 weeks apart
- Booster 1 - 2 weeks after initial immunization
- Booster in 1 year and then every 1 - 3 years

CANINE PARVOVIRUS

Pathogen

Reservoir: dogs
Small non-enveloped DNA virus
Stable in environment

Transmission

- Oral
  - Oronasal exposure with CPV-contaminated feces
  - Shed for 7 - 7 weeks
- Fomites
  - Environment for months

Pathophysiology

Replicates into lymphoid tissue of oropharynx and mesenteric LN
Viremia spreads to all rapidly dividing cells throughout body, including bone marrow
Replicated in germinal epithelium of intestinal crypts destroying epithelium
Bacteria translocate in blood - sepsis and coagulation disorders
Replicates in myocardium causing myocarditis and sudden death
Replicates in myocardium if infected in utero or before 8 weeks old
Shed in feces for 10 days

Clinical Signs

- GI signs: vomiting, diarrhea, inappetence
  - Hemorrhagic, profuse and has fetid smell
- Fever
- Abdominal pain
- Dehydration
- Leukopenia - weakness, seizures
- Cardiomyopathy
- Sepsis
  - Neonatal disease

Diagnostics

- CBC: leukopenia, lymphopenia, thrombocytopenia
- Cytology: inclusion bodies in lymphoblasts
- **Fecal ELISA antigen test**
  - False negative: lag early in 10 days post infection
  - False positive: 4 - 5 days after MLV vaccine

Treatment

- Canine Parvovirus Monoclonal Antibody
- Supportive care 72 - 96 hours
  - IV fluids, K+, dextrose
  - Antibiotics for secondary bacterial infection
  - Antiemetics
  - Antacids
  - Antiparasitic
  - Nutritional therapy - early
  - Plasma therapy
- Outpatient: SQ fluids, antiemetics, antimicrobials, syringe feeding

Prognosis

Patients that survive the first 3 - 4 days typically recover
Fatal if develop hypoglycemia
Worse prognosis with severe lymphopenia or coagulopathies

Disinfection

- Very stable and ubiquitous in environment
- Bleach 30 minutes contact
  - Other organic stable removal
- 30 minutes contact time
- Quaternary NH4
  - Accelerated hydrogen peroxide
  - Organic debris removal
  - 1 minute contact time

Vaccination

- Modified Live Virus - DO
  - DAPP or DHPP
  - Same schedule as distemper shot
- Recombinant Canine Adenovirus-2 from poxvirus can transfer a high level of maternal antibodies to the puppies
  - Not affected by maternal antibodies until 5 months
  - Vaccinate until 20 weeks

CANINE INFECTIOUS HEPATITIS

Pathogen

Canine adenovirus type 1
Stable in environment

Transmission

- Oronasal exposure of CAV-1 infected bodily fluids
  - Tissue, urine, saliva
  - Fomites

Pathophysiology

- Infection of tonsilar and oropharyngeal exposure
  - Spreads to all secretions
- Viremia within 4 - 8 days
- Shed in secretions
  - Vascular endothelial cells
  - Hepatic cells
- Immune response controlled by day 7 if clear virus
- Persist in kidneys, cornea, bronchus and hepatic fibrosis

Clinical Signs

Bloody nose, anorexia, lethargy
Vomiting, diarrhea
Hepatitis
Glaucoma, blue cornea, corneal edema
Hepatic dysfunction: Hepatomegaly, coagulopathies, icterus
Renal disease: nephropathy, azotemia, anuria, dysmotility
Sudden death

Diagnostics

CBC: leukopenia, thrombocytopenia, anemia
Chemistry: ALT > AST elevated bilirubin
Virus isolation: can allow diagnosis

Biopsy of liver, viral inclusion bodies, PCR

Treatment

- No specific therapy
- Supportive care
  - IV fluids
  - Hepatic encephalopathy cases
  - Antibiotics
  - Antibiotics if concurrent bacterial infection
- Liver protectants DAMe, ursodiol, vitamin D

Prognosis

- Recovery 90 - 95%
  - Hepatic encephalopathy present
- Hepatic failure from hepatic necrosis
- Chronic active hepatitis
- Hepatic fibrosis

Prevention / Vaccination

- Resistant to many disinfectants: bleach, steam cleaning
- Modified live virus vaccine
  - Associated with immune-mediated "blue eye" and renal disease
  - Induce viral shedding
  - Recombinant vaccine is free of canine infectious tracheobronchitis
- CAV-2 vaccine
  - Antibodies protect against CAV-1 and CAV-2
  - Titer > 1:32 - 1:64 protect against CAV-1 and CAV-2
- Modified live Rabies - DO
  - Same schedule as distemper
  - K + 1 - 52
  - Same schedule as distemper

INFECTIOUS TRACHEOBRONCHITIS

Pathogen

- Canine Infectious Respiratory Disease (CIRD)
- Canine Respiratory Disease Complex
- Kennel Cough
- Virus
  - Parainfluenza (PI)
  - Adenovirus-2
  - Herpesvirus
  - Coronavirus
- Bacteria
  - Bordetella bronchiseptica (Bb)
  - Mycoplasma

Transmission

- Reservoir: dogs
- Most common mycoplasma: Bordetella, CAV-2, CDV, CPV
- Aerosolized droplets, contact, fomites
- Direct dog-to-dog contact
- Fomites
- Inhalation in respiratory epithelium
- Incubation period: 3 - 7 days
- Shed for up to 14 days

Clinical Signs

- Dry hacking cough - "honking cough"
  - Gagging, retching, vomiting
  - May end in retching or gagging
- Rhinitis: Tracheitis, clear nasal discharge
- Mucopurulent discharge
- Enlarged cervical lymphadenopathy
  - Moist productive cough
  - Dyspnea
  - Dyspnea (pneumonia)

Diagnostics

- Pneumonitis in most cases based on clinical signs and history
- Rule out other causes of pneumonitis and bronchitis
- Pathogen identification is unnecessary in most cases
  - Recommended if
    - Severe cases or suspicious case
    - Systemic disease present
    - Outbreak in multiple dogs
  - **PCR of nasal or oropharyngeal swabs**

Treatment

- Uncomplicated
  - Supportive care
    - Cough suppressant
    - Highly palatable food
    - Nebulization of humidifier
  - Antibiotics
  - Doxycycline
    - Flavor
    - Discharge
- Complicated
  - Antibiotics
  - Fluid therapy
  - Oxygenation
  - Coagulation

Prognosis

- Uncomplicated very good
  - Recovery 7 - 10 days
  - Recovery 6 - 10 days without treatment
  - Secondary pneumonia is good

Vaccination

- Parainfluenza, CAV-2, CPV
  - Modified live virus - DO
  - Protects against clinical signs but doesn't prevent infection or viral shedding
- Parainfluenza, CAV-2, Bordetella
  - Intranasal, oral, subcutaneous - intranasal, Oral
  - Protects against clinical signs, prevents infection and viral shedding
- Bordetella bronchiseptica bacterin - intranasal
  - Protects against clinical signs, prevents infection and viral shedding
  - Recombinant - oral
  - Protects against clinical signs, prevents infection and viral shedding
- Inactivated cellular antigen extract - DO
  - Protects against clinical signs, but doesn't prevent infection or viral shedding

Vaccine Schedule

- Kick-off puppy - 8 weeks
  - Same schedule as distemper shot
- Bordetella bronchiseptica bacterin
  - Initial - 2 days only
  - Booster every 6 months in high risk environments
  - Booster every 12 months for low risk or at - 2 weeks
- K/3 Bordetella
  - Initial 2 Weeks 3 - 4 weeks apart
  - Booster every 6 months in high risk environments

LEPTOSPIROSIS

Pathogen

- Gram negative spirochete
  - Pathogen serovars of L. interrogans
  - Canicola
  - Icterohaemorrhagiae
  - Pomona
  - Grippotyphosa
  - Bratislava
- Public work, hunt, athletic environments
- Stagnant or slow moving water
- Wildlife, farm animals in rural environments
- Zoonotic

Transmission

- Reservoir: skunks, raccoons, opossums, pigs
  - Shed in urine for months-years
  - Leptospirosis/Hepatitis - cat
  - Pomona - cow, pig
  - Grippotyphosa - voles, muskrats, skunks
  - Bratislava - cat, pig, horse
- Contact with infected urine
- Ingestion of tissues
- Indirect: exposure to water sources, soil, food, or bedding contaminated with infected urine
  - Free residence is lost - November

Pathophysiology

Gram-through, thrombocytopenia, or hepatic skin
Multiply quickly once in blood stream
Hemolysis, hemorrhage, vasculitis, thrombocytopenia, hemorrhage
Spread to and replicate in
Kidneys - tubules - shed for weeks to months
Liver - cholestasis, hepatocellular necrosis, hepatitis - jaundice
Lungs - vasculitis and hypoxic injury - capillaritis/ pulmonary hemorrhage syndrome
Spleen, Eyes, CNS
Incubation period: 7 days

Clinical Signs

- Fever
- Myalgia, muscle tenderness and death with few signs
- Acute
  - Vomiting, fever, lethargy, anorexia, diarrhea
  - Acute kidney failure - PU/PD, Oliguria
  - Liver failure - icterus, hemorrhage
  - Lung - cough, dyspnea
  - Mesh cases of pulse, large spots are 48 hr age
- Subclinical
  - Anorexia, vomiting, polydipsia, weight loss, uveitis
  - Uveitis, hemorrhagic signs
- Chronic
  - Chronic active hepatitis
  - Chronic kidney failure

Diagnostics

- CBC: leukocytosis, anemia, thrombocytopenia
- Chemistry: azotemia, hyperbilirubinemia, increased ALT, ALT, bilirubin
- Urine: hyperbilirubinuria, granular casts
  - Leptospira with special staining or dark field microscopy
- **Serology: Microscopic Agglutination Test**
  - False negative: first 7 - 10 days of acute infection
  - False positive: vaccination, need paired titers
- **PCR on blood, urine**
  - Gram Polish of urine test SATRIEC, __ agglu, __ and Shield __ agglu

Treatment

- Antibiotics
  - Doxycycline, do not eliminate renal carrier state
  - Is fluoroquinolone does eliminate renal carrier state
- Supportive care
- Antiemetics
- Fluid therapy
- Liver support: hepatoprotectants (SAMe, ursodiol)

Prognosis

- Recovery with aggressive fluid therapy and antibiotics: good
- Survival rate 90 - 98%
- Poor prognosis: pulmonary disease with Pomona
- Chronic kidney disease: Grippotyphosa, Bratislava
- Chronic active hepatitis (cholestasis/hepatitis): Canicola/Icterohaemorrhagiae

Prevention / Vaccination

- 2 doses 2 weeks to adequate water
- Decrease access to stagnant water
- Rodent control
  - 2 Canine to rodent survival
  - Leptospira Interrogans 4 wav - 1 - Canicola, 1, pomona, 1, grippotyphosa, 1, icterohemorrhagiae

Vaccine Schedule

- Leptospirosis
  - Initial - 2 doses given 2 weeks of age
  - Booster 3 - 4 weeks later
  - Annual booster recommended
- K/3
  - Given with CAVP at 2 and 16 weeks
  - Annual booster recommended

CANINE INFLUENZA VIRUS

Canine Influenza: Pathogen

- H3N2 canine influenza
  - H3N8 - equine
  - H1N2
- Enveloped, ssRNA virus
- Host specific

Transmission

- Aerosol transmission
- Fomites
- Direct oronasal contact
- Shed in secretions and feces
- Replication in mucosal epithelium of respiratory tract
  - H3N2 shed 20 days
- Incubation 2 - 4 days

Clinical Signs

Mild to severe, depending, mucopurulent nasal discharge
Tracheitis, cough
Lethargy
Dehydration
Secondary bacterial pneumonia (pH rare)
Fulminate death, Disseminate Thrombocytopenia failure
Acute Respiratory Distress Syndrome

Diagnostics

- Pneumonitis in most cases based on clinical signs and history
- Pathogen identification is unnecessary in most cases
  - Recommended if
    - Severe cases or suspicious case
    - Systemic disease present
    - Outbreak in multiple dogs
  - **PCR, virus isolation, and ELISA of nasal or oropharyngeal swabs**

Treatment

- No specific therapy
- Supportive care
  - Cough suppressants
  - Fluid therapy
  - Oxygen
  - Nebulization
  - Nutritional support
- Antibiotics
  - Oral
  - Discharge
  - Pneumonia

Prognosis

Most self-limiting and produces only mild clinical signs
Occasional death

Prevention / Vaccination

- Sensitive to routine disinfectants
- Vaccine - DO
  - H3N8, H3N2, or H3N2/H3N8 bivalent
  - Decreases severity of signs and viral shedding
  - Initial 2 doses 2 - 4 weeks apart
  - Annual vaccine boosters